

counter and only with difficulty (making them illiquid). Over the counter markets have *search frictions*, including finding a counterparty and overcoming the informational disadvantages an investor might have in evaluating the underlying value of portfolio companies. Even launching a PE fund entails search friction, since asset owners and potential GPs must find and evaluate one another. Transactions costs in secondary markets for PE are enormous: Harvard University, for example, faced discounts of 50% trying to exit from PE funds during the financial crisis (see chapter 11).⁵

The valuation of public equity is objective, happens immediately, and is easily verifiable through market pricing. PE, on the other hand, is hard to value. The valuation process is infrequent, subjective, and (unnecessarily and mysteriously) complex. Most PE funds have a stated horizon of around ten years, but many extend beyond, sometimes past fifteen. Thus, a true return on a PE fund cannot be computed for a decade or more, when liquidation finally occurs. The long horizon of PE means that investors cannot easily rebalance PE positions, which renders standard (usually mean-variance) portfolio choice models inappropriate, as discussed in chapter 13.

Finally, contracts for liquid assets are largely standardized and transparent. PE contracts are anything but, requiring far more effort and diligence from investors.

2.3. TYPES OF PRIVATE EQUITY

There are several types of PE:⁶

Leveraged buyout (or simply buyout) funds buy mature companies, often already listed, using high levels of debt. (The average leverage ratio of buyout funds is three.⁷) This subclass represents most of the money allocated to PE, and sometimes the term “PE” is used to refer to buyout funds. In this chapter, I take a broader view of PE.

Venture capital (VC) funds take stakes in companies at early stages of development using little or no leverage. *Angel investors* are special types of VC investors and typically use only their own money, rather than relying on a pool of investors as do VC funds. Angels specialize in the earliest stage of development—the proverbial startup in someone’s garage. Apple, Google (both of which actually did start in garages), Amazon, and YouTube were all financed this way. That these are high-tech companies is no accident—VCs gravitate to high-growth startups, although some VC firms specialize in other areas.

PE *real estate* funds invest in property as well as a real estate development. Real estate funds tend to be highly levered.

⁵ See “Liquidating Harvard,” Columbia CaseWorks ID #100312.

⁶ Most academic research focuses on buyout and VC funds because of lack of data in other subclasses.

⁷ See Axelson et al. (2012).